

Identifying Post-COVID Risk Factors with Model-Agnostic Feature Importance

using the `xplainfi` R package on the German national cohort (NAKO)

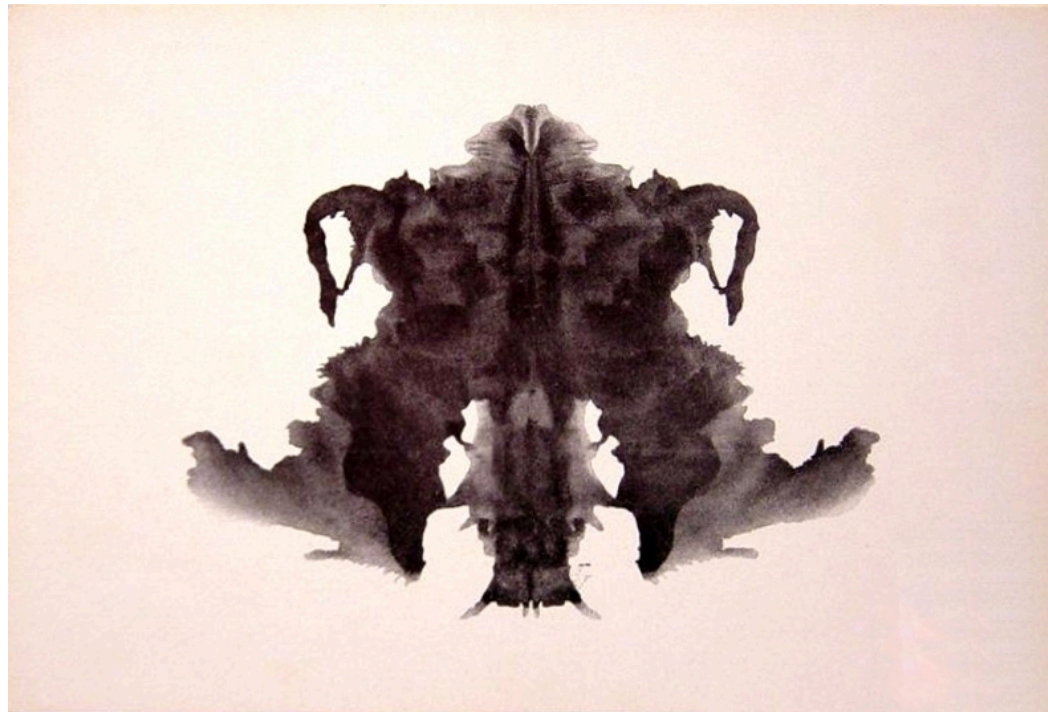
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CEN 2026

A matter of interpretation



Post-COVID condition (PCC)

“Long- / Post-COVID”



2

- Persistent / recurring symptoms long after SARS-CoV-2 infection
- Clinically heterogenous (symptoms & severity)

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 - “Any” PCC: at least **1**, “severe” PCC: at least **9**
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(Federal Ministry of Research, Technology and Space)

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- **FI $\neq$ causal effect** but associations still informative (Ewald et al., 2024)
- “Importance” itself is not a single quantity → *method choice* defines the answer

Common FI methods

How “important” is feature X_j ?



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Leave-one-covariate-out (*LOCO*)

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Permute, but respect feature dependence



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- CFI: perturbation *conditional* on other features
 - PFI: $\tilde{X}_j \sim F_{X_j}$, with $\tilde{X}_j \perp\!\!\!\perp (X_{-j}, Y)$
 - CFI: $\tilde{X}_j \sim F_{X_j | X_{-j}}$, with $\tilde{X}_j \perp\!\!\!\perp Y \mid X_{-j}$

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- Requires *conditional sampling*, some options:
 - Conditional Gaussian $\rightarrow$ fast, but only continuous data
 - Adversarial random forests (ARF) $\rightarrow$ handles mixed data, computationally heavier

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Fixed model, learning algorithm, data-generating process?

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- Estimand should match the research question
 - “Why does *this* model predict X?” → model-level
 - “Which features does *this prediction algorithm* rely on?” → learner-level

Analysis dataset: NAKO

German National Cohort



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- $N \approx 66250$ participants reporting ≥ 1 SARS-CoV-2 infection
- $\approx 24\%$ classified as **“any PCC”** (“Severe PCC” much rarer $\rightarrow$ not shown today)

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 - comorbidities, smoking history, biomarker labs (where available)
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 - demographics, anthropometrics, socioeconomic
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- ≈ 150 features used (dropping near-constant / near-fully-missing)
- Rare-event indicators retained despite low prevalence

Analysis: Any PCC prediction

Preliminary, exploratory



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 - → general performance **42–44% PR-AUC** (baseline = 24%)
- PFI, CFI (+ARF), LOCO computed via `xplainfi`

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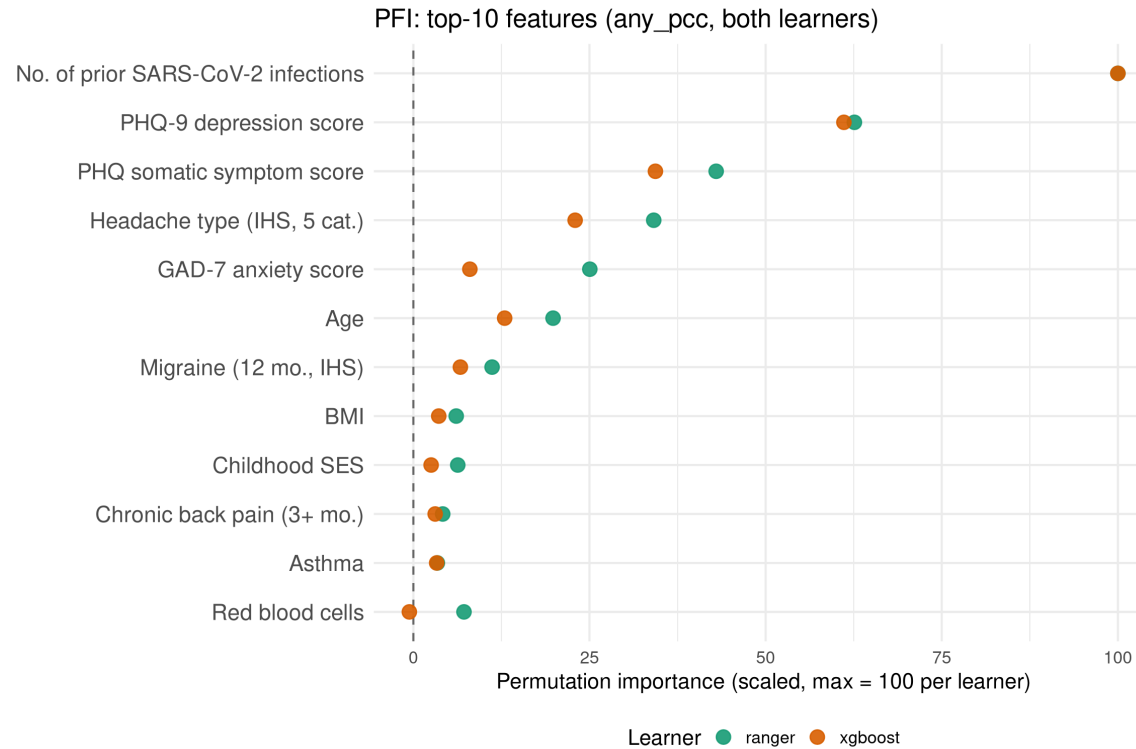


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- PFI, CFI (+ARF), LOCO computed via `xplainfi`
- FI here on test set (i.e. **model importance**)
- FI scores on scale of metric, hard to interpret
 - → Scaled such that 100 = most important feature
 - Noteworthy: Do rankings change between methods, learners?

Results: PFI

Top 10 features

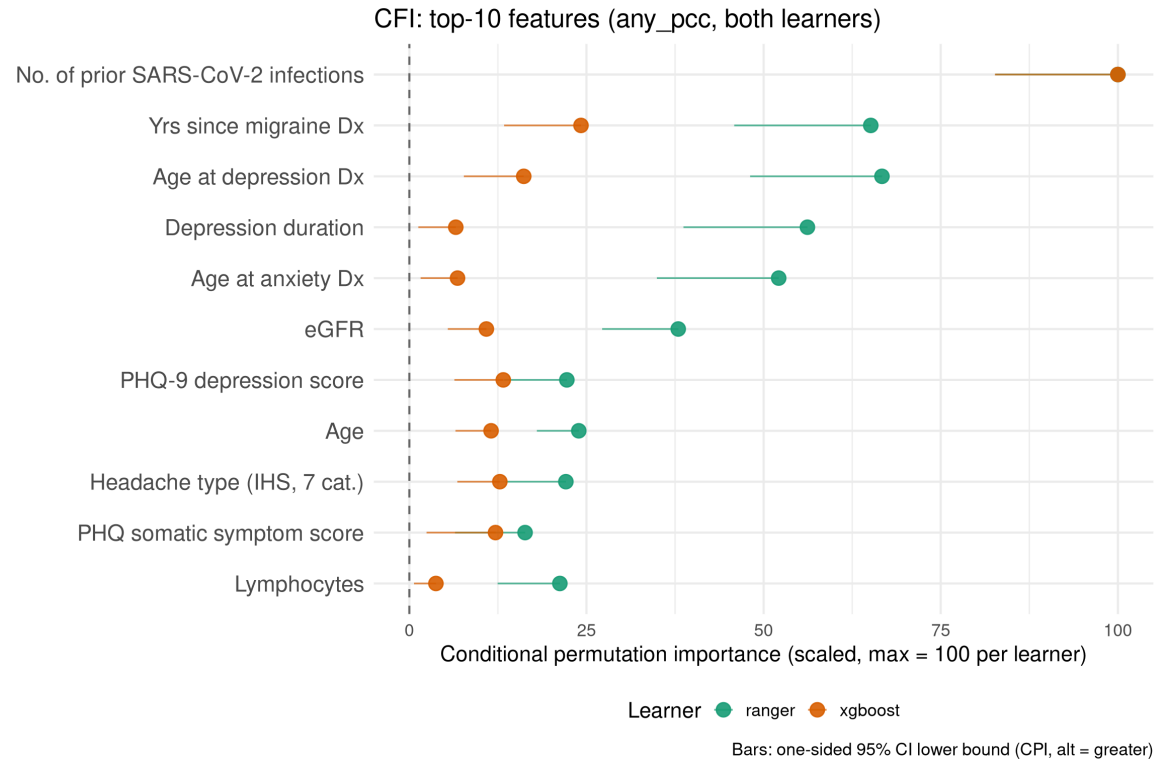


Results: CFI

Top 10 features

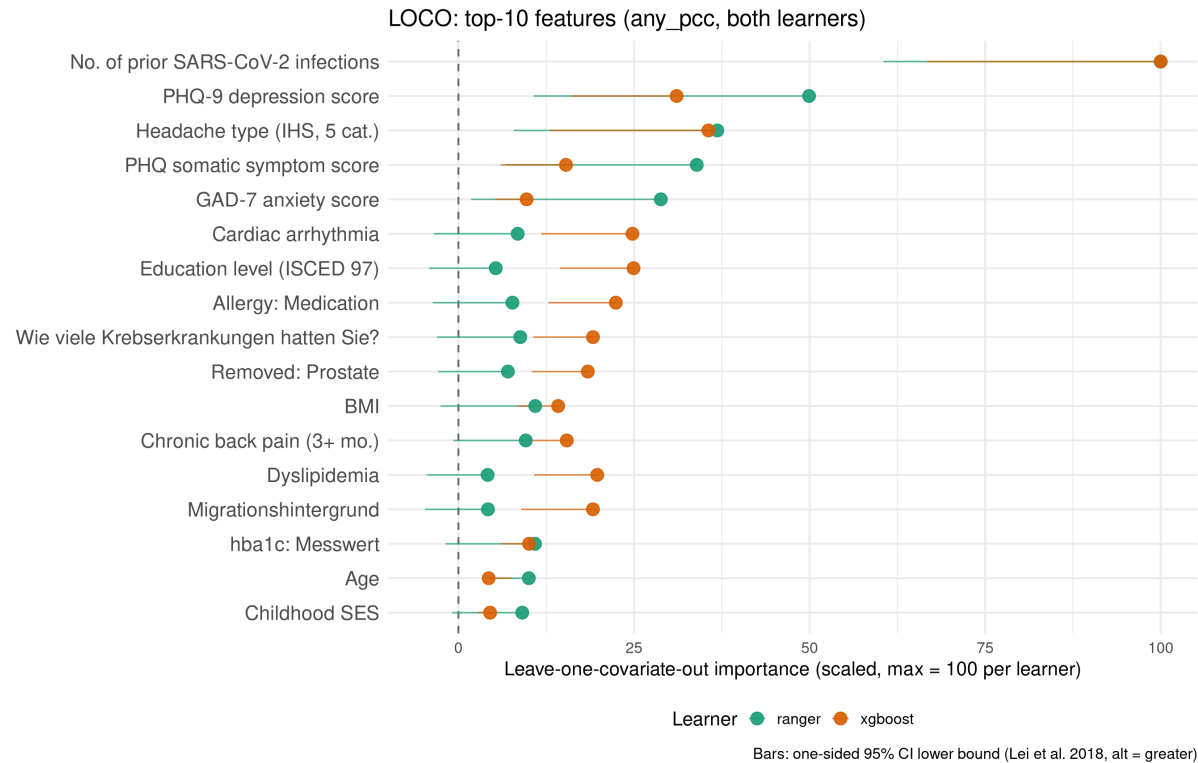


10



Results: LOCO

Top 10 features



Same features, different rankings

What changes between methods?



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- Top-ranked features under PFI not always top under CFI / LOCO
 - CFI *should* de-value importance of correlated features
 - PFI can over-credit features that are easy to extrapolate around

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- Top-ranked features under PFI not always top under CFI / LOCO
 - CFI *should* de-value importance of correlated features
 - PFI can over-credit features that are easy to extrapolate around
- **Ranks are often more interpretable than raw values**
 - Raw PFI/CFI/LOCO are on the scale of the evaluation metrics
 - Magnitudes hard to compare across methods

Cocnclusion

Everything is complicated, always



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- Feature importance is useful, but **not a single number**
- Each method answers a slightly different question
 - PFI: marginal, model-faithful but extrapolation-prone
 - CFI: conditional, more faithful to joint, sampling-dependent
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- Each method answers a slightly different question
 - PFI: marginal, model-faithful but extrapolation-prone
 - CFI: conditional, more faithful to joint, sampling-dependent
 - LOCO: refit-based, expensive but assumption-light
- Match the **estimand** to the **research question**
- Compare methods → robustness check, not contradiction
- `xplainfi` provides a unified interface for all of the above (Burk et al., 2026)

Thank you for your attention!



Contact

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